**What is the causal effect of substantial state minimum-wage increases of at least $0.25 on employment among teenagers ages 16–19?**

## 1. Empirical Approach

### 1.1 Data and Sample

I use the IPUMS Current Population Survey (CPS) Basic Monthly samples from January 2010 through December 2019 (Flood et al. 2025). I restrict the raw CPS sample to civilian respondents ages 16 through 19. The CPS provides each respondent's state of residence, age, sex, employment status, and Final Basic Weight (WTFINL). I use all twelve monthly samples in each year to increase the number of teen observations available within each state and month.

I transform the individual CPS records into a state-by-month panel before estimation. For each state-month, I use WTFINL to calculate the survey-weighted teen employment rate and the weighted shares of respondents who are female and ages 17, 18, and 19. Age 16 is the omitted age category in models that include demographic composition controls. I then merge these monthly outcomes with state minimum-wage data from Vaghul and Zipperer (2022) by state, year, and month. The minimum-wage series records the effective wage floor, meaning the higher applicable state or federal minimum wage.

The treatment definition responds to the concern that a simple treated-versus-untreated indicator would treat very small cost-of-living adjustments the same as larger policy changes. I define a substantial increase as a rise in the effective minimum wage of at least $0.25. Cengiz et al. (2019) use the same dollar threshold as part of their definition of prominent minimum-wage events, although their definition also requires that at least 2 percent of workers are directly affected. I use the $0.25 threshold as guidance but do not call my events "prominent" because the Basic Monthly CPS extract used here does not contain the hourly wage information needed to reproduce their worker-exposure criterion.

I organize the data as stacked difference-in-differences sub-experiments. A focal event must have no other increase of at least $0.25 during the prior 24 months, which prevents annual phase-in steps from automatically becoming separate new events. I also require no additional increase of at least $0.25 during the following 11 months. For each qualifying event, I keep a balanced window from 12 months before treatment through 11 months after treatment. The comparison group for each cohort consists of states with no increase of at least $0.25 anywhere in that same 24-month event window. A state may serve as a clean control in one cohort and appear in another cohort if it satisfies the relevant event-window rules.

The final stacked sample contains 33 treated state-events in 11 treatment cohorts and 10,008 state-month observations drawn from all 50 states and the District of Columbia. The data contain 417 state-by-cohort units: 33 treated units and 384 control units. Because control states can contribute to more than one cohort, the 10,008 observations are stacked state-month observations rather than 10,008 unique state-months.

### 1.2 Measures

The dependent variable is the state-month teen employment rate. It is the survey-weighted share of civilian respondents ages 16–19 whom the CPS classifies as “at work” or “has job, not at work last week.” Teen employment is a standard outcome in minimum-wage research; Allegretto et al. (2011) and Giuliano (2013) also study how minimum-wage changes relate to teen employment. I express the outcome as a proportion in the regressions and convert coefficients to percentage points for reporting.

The main explanatory variable is Treated × Post. Within each stacked sub-experiment, Treated identifies the state or states experiencing the focal minimum-wage increase, and Post equals one from the event month through month +11.

Their interaction therefore measures the change in teen employment in treated states after the increase relative to clean control states in the same cohort. Model 2 additionally controls for the state-month female share and the shares ages 17, 18, and 19. These variables account for short-run changes in the demographic composition of the teen sample; age 16 is the omitted category.

### 1.3 Descriptive Statistics

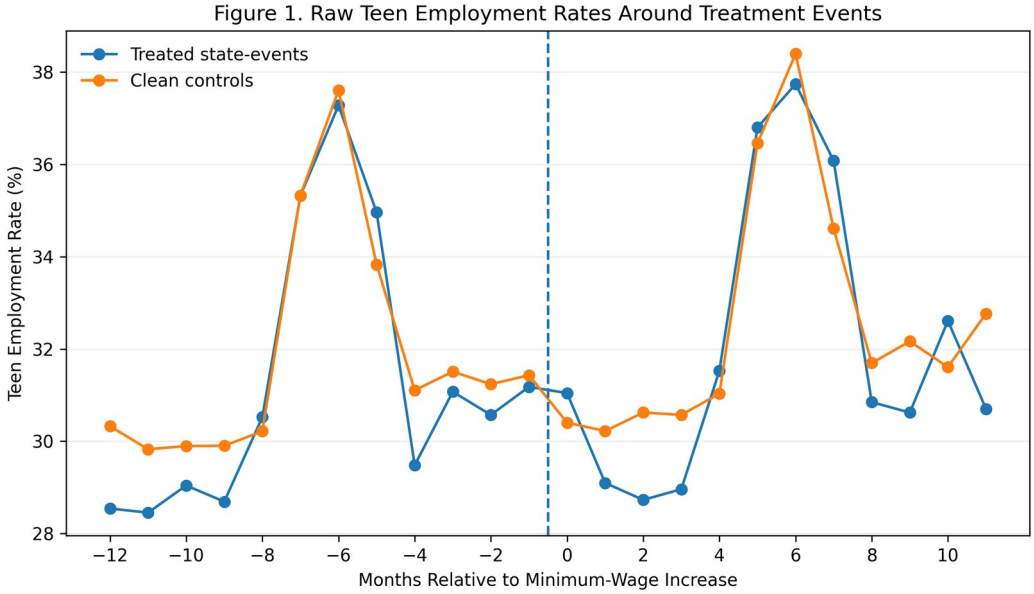
**Table 1. Descriptive Statistics**

| Variable | Full stacked sample | Treated stacks | Control stacks |
|---|---|---|---|
| Teen employment rate (%) | 32.16 | 31.66 | 32.20 |
| Female share (%) | 49.24 | 48.85 | 49.27 |
| Average age (years) | 17.42 | 17.45 | 17.42 |
| Effective minimum wage ($) | 7.58 | 8.19 | 7.53 |
| State-month observations | 10,008 | 792 | 9,216 |
| State × cohort units | 417 | 33 | 384 |

Notes: Means and percentages use the corrective stacked-DiD weights used in estimation; observation and unit counts are unweighted. “Treated stacks” are focal treated states in their event cohorts. “Control stacks” are cohort-specific clean controls with no minimum-wage increase of at least $0.25 in the same event window. The sample covers event time -12 through +11. CPS state-month outcomes are survey-weighted using WTFINL. Minimum-wage data are from Vaghul and Zipperer (2022).

Table 1 shows that the weighted teen employment rate is 32.2 percent in the final stacked sample. Treated state-events have a slightly lower raw employment rate than controls, but the groups are very similar in average age and gender composition. The effective minimum wage is higher in treated stacks because those states experience the focal minimum-wage increases. These raw differences do not identify a causal effect; the econometric model compares within-state changes around treatment to changes in clean control states.

Figure 1. Raw Teen Employment Rates Around Treatment Events




Notes: Survey-weighted state-month employment rates in the final stacked analytic sample. The dashed vertical line separates the pre-treatment period from the event month and post-treatment period. Series are descriptive and do not include fixed effects or other regression adjustments.

Figure 1 shows substantial month-to-month movement in teen employment, including common seasonal patterns in the treated and control series. The two series often move in the same direction, but the figure is descriptive and cannot establish parallel trends or a treatment effect because it does not remove persistent state differences or cohort-specific calendar-time shocks. I therefore use the stacked DiD model and event-study estimates below for causal inference.

### 1.4 Econometric Model

I use a weighted stacked difference-in-differences design. This approach is appropriate because minimum-wage changes occur in different states at different times and because some states experience multiple increases during the sample period. A first-treatment-only design would compress the policy history into a single adoption date. Instead, the stacked design creates a separate sub-experiment around each qualifying minimum-wage event and compares the treated state or states with clean controls observed over the same event window. This event-based

logic is similar to the minimum-wage design in Cengiz et al. (2019) and follows recent guidance for staggered DiD designs (Wing et al. 2024).

A basic stacked estimator can give treated and control observations different implicit weights when cohorts contain different numbers of treated and control units. Wing, Freedman, and Hollingsworth (2024) show that corrective sample weights fix this problem. I therefore use corrective stacked-DiD weights so that each cohort's controls receive weight consistent with that cohort's share of treated state-events. Each sub-experiment uses a balanced 24-month event window, from month -12 through month +11.

The main regression is:

$$Employment_{sct} = \alpha_{sc} + \lambda_{ct} + \beta(Treated_{sc} \times Post_{ct}) + X'_{sct}\,\gamma + \varepsilon_{sct}. \quad (1)$$

In Equation (1), Employment is the survey-weighted teen employment rate for state s in cohort c and calendar month t. The coefficient of interest, β, measures the average change in teen employment after a substantial minimum-wage increase in treated states relative to clean controls. The state-by-cohort fixed effects, α_sc, absorb persistent differences between each state and the comparison states within a given sub-experiment. The calendar-month-by-cohort fixed effects, λ_ct, absorb shocks common to states within the same cohort in the same calendar month. X contains the demographic composition controls used in Model 2. Model 1 omits X and is the preferred simpler specification; Model 2 checks whether the estimate changes when female and age shares are included.

The key identifying assumption is parallel trends: absent the minimum-wage increase, teen employment in treated states would have evolved similarly to teen employment in the clean control states within the same cohort. I assess this assumption with an event-study specification that replaces Treated × Post with monthly treatment indicators from event month -12 through +11 and omits month -1 as the reference period. The pre-treatment coefficients from months -12

through -2 show whether treated states were already moving differently before the policy change.

Because treatment varies at the state level, I cluster conventional standard errors by state. However, with only 51 state-level clusters, I do not rely only on standard large-sample clustered inference. I also report wild cluster bootstrap p-values using 9,999 replications, following the small-cluster inference approach implemented by Roodman et al. (2019). I use the wild-bootstrap p-values as the primary significance check.

## 2. Results

Table 2 reports the main stacked DiD estimates. In Model 1, the Treated × Post coefficient is 0.109 percentage point, with a state-clustered standard error of 0.499 percentage point and a wild cluster bootstrap p-value of 0.830. Because the coefficient is not statistically significant, I do not interpret its positive sign or magnitude as evidence that minimum-wage increases raise teen employment. The conventional 95 percent confidence interval includes both negative and positive values. The main result is therefore that the analysis does not detect a short-run effect on teen employment; it does not prove that the true effect is exactly zero.

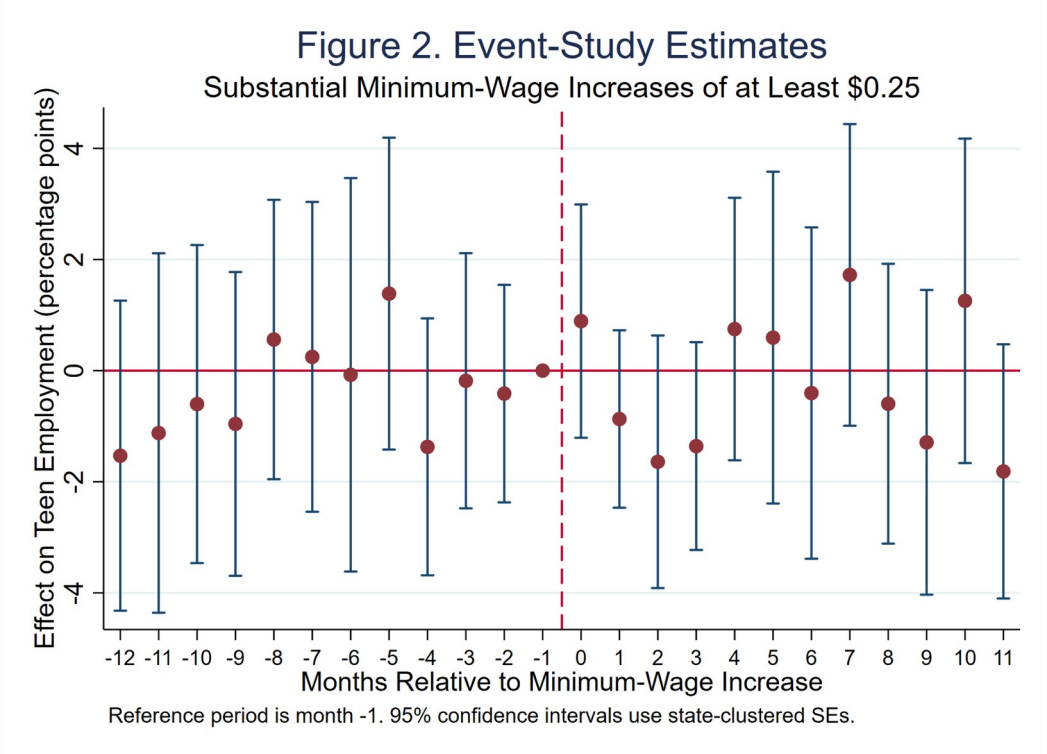
Model 2 adds the female share and age composition controls. Its Treated × Post coefficient is 0.106 percentage point, with a wild cluster bootstrap p-value of 0.836, so it is also not statistically significant. The coefficient changes very little after adding the demographic controls. This stability suggests that short-run changes in the measured age and gender composition of the teen sample are not driving the main null result.

**Table 2. Stacked Difference-in-Differences Estimates of Teen Employment**

| | Model 1: Main | Model 2: + demographic controls |
|---|---|---|
| Treated × Post coefficient (percentage points) | 0.109 | 0.106 |
| State-clustered SE (percentage points) | 0.499 | 0.507 |
| Conventional clustered p-value | 0.829 | 0.835 |
| Wild-cluster-bootstrap p-value | 0.830 | 0.836 |
| Demographic composition controls | No | Yes |
| State × cohort fixed effects | Yes | Yes |
| Calendar month × cohort fixed effects | Yes | Yes |
| Corrective stacked-DiD weights | Yes | Yes |
| Observations | 10,008 | 10,008 |
| State clusters | 51 | 51 |

Notes: Dependent variable is the survey-weighted state-month teen employment rate. Coefficients and standard errors are reported in percentage points. Both models use the final clean stacked sample, corrective stack weights, state-by-cohort fixed effects, and calendar-month-by-cohort fixed effects. Conventional standard errors are clustered by state. Wild-cluster-bootstrap p-values use 9,999 replications clustered by state. Model 2 adds state-month female share and age 17, 18, and 19 shares; age 16 is the omitted age category.

Figure 2 reports the event-study estimates. The monthly pre-treatment coefficients fluctuate around zero and their 95 percent state-clustered confidence intervals all include zero. A conventional joint test of the eleven pre-treatment coefficients gives $p = 0.085$, while the wild cluster bootstrap joint test gives $p = 0.410$. Because the wild bootstrap is the preferred inference procedure in this setting, I do not reject the null that the pre-treatment coefficients are jointly zero. This result is consistent with, although it cannot prove, the parallel-trends assumption.




Notes: Monthly stacked-DiD event-study coefficients for substantial minimum-wage increases of at least $0.25. Event month -1 is the omitted reference period. Bars show 95 percent confidence intervals based on state-clustered standard errors. The wild cluster bootstrap joint test for all pre-treatment coefficients from month -12 through -2 has p = 0.410.

The post-treatment event-study estimates also move above and below zero, and none of their 95 percent clustered confidence intervals excludes zero. There is no persistent downward pattern after treatment. Taken together with the average DiD estimate, the results provide no evidence that the substantial minimum-wage increases in this sample caused a detectable decline in teen employment during the following twelve months.

The results have several limitations. First, the design estimates short-run effects around clean increases of at least $0.25, so it does not identify the effects of very small cost-of-living adjustments, longer-run cumulative phase-ins, or every type of minimum-wage policy. Second, I use the $0.25 threshold from the Cengiz et al. framework but cannot reproduce their additional requirement that at

least 2 percent of workers are directly affected because the Basic Monthly CPS extract does not contain the hourly wage distribution needed for that calculation. Third, CPS state-month teen employment rates come from survey samples and can be noisy, especially in smaller states. Finally, parallel trends cannot be observed directly. The event-study results are reassuring, but they cannot rule out every unobserved state-specific shock that happens near treatment.

## 3. References


Allegretto, Sylvia A., Arindrajit Dube, and Michael Reich. 2011. "Do minimum wages really reduce teen employment? Accounting for heterogeneity and selectivity in state panel data." *Industrial Relations* 50 (2): 205–240. https://doi.org/10.1111/j.1468-232X.2011.00634.x.

Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer. 2019. "The effect of minimum wages on low-wage jobs." *Quarterly Journal of Economics* 134 (3): 1405–1454. https://doi.org/10.1093/qje/qjz014.

Flood, Sarah, Miriam King, Renae Rodgers, Steven Ruggles, J. Robert Warren, Daniel Backman, Etienne Breton, Grace Cooper, Julia A. Rivera Drew, Stephanie Richards, David Van Riper, and Kari C. W. Williams. 2025. *IPUMS CPS: Version 13.0 [dataset]*. Minneapolis, MN: IPUMS. https://doi.org/10.18128/D030.V13.0.

Giuliano, Laura. 2013. "Minimum wage effects on employment, substitution, and the teenage labor supply: Evidence from personnel data." *Journal of Labor Economics* 31 (1): 155–194. https://doi.org/10.1086/666921.

Roodman, David, James G. MacKinnon, Morten Ørregaard Nielsen, and Matthew D. Webb. 2019. "Fast and wild: Bootstrap inference in Stata using boottest." *Stata Journal* 19 (1): 4–60. https://doi.org/10.1177/1536867X19830877.

Vaghul, Kavya, and Ben Zipperer. 2022. "Historical state and sub-state minimum wages." Version 1.4.0. https://github.com/benzipperer/historicalminwage/releases/tag/v1.4.0.

Wing, Coady, Seth M. Freedman, and Alex Hollingsworth. 2024. "Stacked difference-in-differences." NBER Working Paper 32054. https://doi.org/10.3386/w32054.

Wing, Coady, Madeline Yozwiak, Alex Hollingsworth, Seth Freedman, and Kosali Simon. 2024. "Designing difference-in-difference studies with staggered treatment adoption: Key concepts and practical guidelines." *Annual Review of Public Health* 45: 485–505. https://doi.org/10.1146/annurev-publhealth-061022-050825.